



FORT STREET HIGH SCHOOL

Year 12 Mathematics Extension 1

Task 1

2024

Student Name:

Student Number:

Teacher	Stamoulos	Razzaghi	Moon	Kaur	Hussein
	☐	☐	☐	☐	☐

General Instructions

- Reading time – 5 minutes
- Working time – 75 minutes
- Write using black pen.
- Board-approved calculators may be used.
- A reference sheet is provided.
- Start each question in the correct answer booklet.
- For all questions show relevant mathematical reasoning and/or calculations.
- Marks may be deducted for poorly set work.

Question	Total	Marks
MC	5	
Q6	8	
Q7	6	
Q8	7	
Q9	7	
Q10	7	
Total	40	
	Percent	

Section I

5 marks

Attempt Questions 1-5

Allow about 8 minutes for this section

Use the multiple choice answer sheet for Questions 1-5.

1 Which of the following is the co-efficient of x^4 in the expansion of $(2 - 5x)^7$?

(A) ${}^7C_4(2)^3(5)^4$

(B) $-{}^7C_4(2)^3(5)^4$

(C) ${}^7C_4(2)^4(5)^3$

(D) $-{}^7C_4(2)^4(5)^3$

2 Given that $\underline{u} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$ and $\underline{v} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$, what is the value of $|\underline{u} - \underline{v}|$?

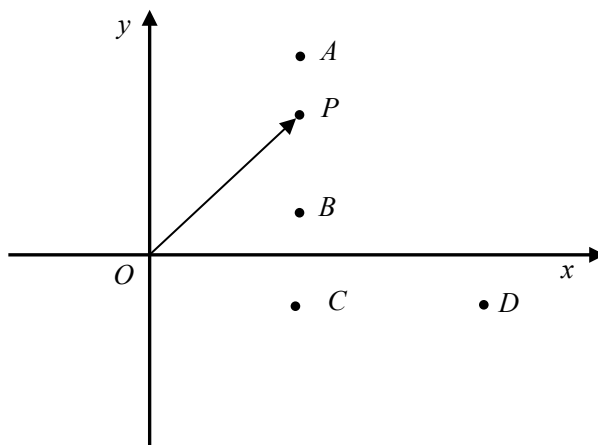
(A) $\sqrt{40}$

(B) 6

(C) $\sqrt{18}$

(D) 2

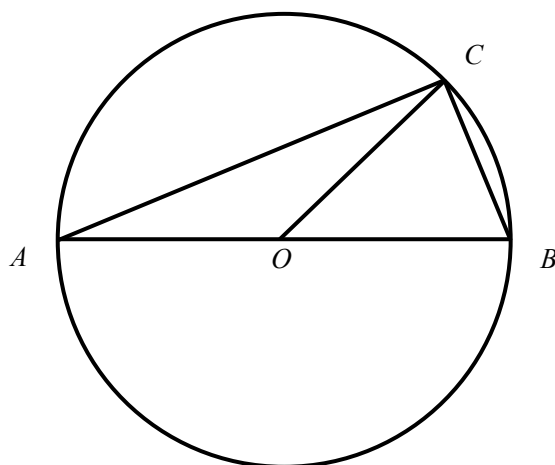
- 3 The diagram below shows the points A , B , C , D and P and the vector $\overrightarrow{OP}$.



Which of the following has the greatest magnitude?

- (A) $\text{proj}_{\overrightarrow{OA}} \overrightarrow{OP}$
- (B) $\text{proj}_{\overrightarrow{OB}} \overrightarrow{OP}$
- (C) $\text{proj}_{\overrightarrow{OC}} \overrightarrow{OP}$
- (D) $\text{proj}_{\overrightarrow{OD}} \overrightarrow{OP}$

- 4 In the diagram below AB is the diameter of the circle with O as its centre. C is a point on the circumference. $\overrightarrow{OC} = \underline{r}$ and $\overrightarrow{BC} = \underline{s}$.



Which of the following is equal to the vector $\overrightarrow{AC}$?

- (A) $\underline{r} - 2\underline{s}$
 - (B) $\underline{r} + 2\underline{s}$
 - (C) $2\underline{r} - \underline{s}$
 - (D) $2\underline{r} + \underline{s}$
- 5 Consider two vectors $\underline{u}$ and $\underline{v}$.

It is known that $|\text{proj}_{\underline{v}} \underline{u}| = |\text{proj}_{\underline{u}} \underline{v}|$ which of the following must always be true?

- (A) $\underline{u}$ is parallel to $\underline{v}$ or $|\underline{u}| \neq |\underline{v}|$
- (B) $\underline{u}$ is parallel to $\underline{v}$ and $|\underline{u}| = |\underline{v}|$
- (C) $\underline{u} \bullet \underline{v} = 0$ or $|\underline{u}| = |\underline{v}|$
- (D) $\underline{u} \bullet \underline{v} = 0$ and $|\underline{u}| = |\underline{v}|$

End of Section I

Section II

35 marks

Attempt Questions 6 - 10

Allow about 67 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available. In questions 6-10, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (8 marks) Use a SEPARATE writing booklet

(a) Given that $\underline{u} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$ and $\underline{v} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

(i) Find the vector $\underline{u} - 3\underline{v}$. 1

(ii) Find $\underline{u} \bullet \underline{v}$. 1

(b) Given that $\underline{u} = \begin{bmatrix} -2 \\ -5 \end{bmatrix}$ and $\underline{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, find the projection of $\underline{u}$ onto $\underline{v}$. 2

(c) Find the angle between the two vectors $\begin{bmatrix} -3 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -7 \end{bmatrix}$. 2

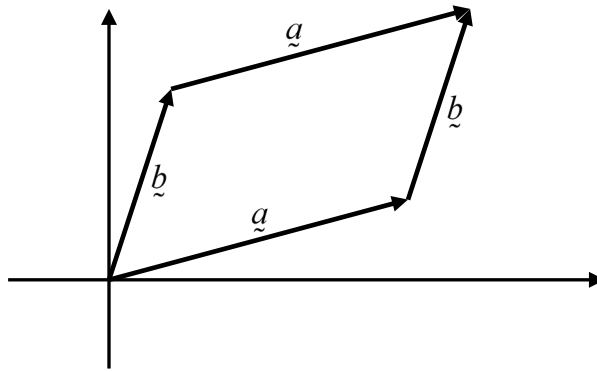
(d) Consider the vectors $\underline{u} = 2\underline{i} - 5\underline{j}$ and $\underline{w} = \underline{i} + b\underline{j}$, where b is real. 2

If $\underline{u} \perp \underline{w}$, find the value of b .

End of Question 6.

Question 7 (6 marks) Use a SEPARATE writing booklet

- (a) Consider the parallelogram formed by the vectors $\underline{a}$ and $\underline{b}$ as shown below.

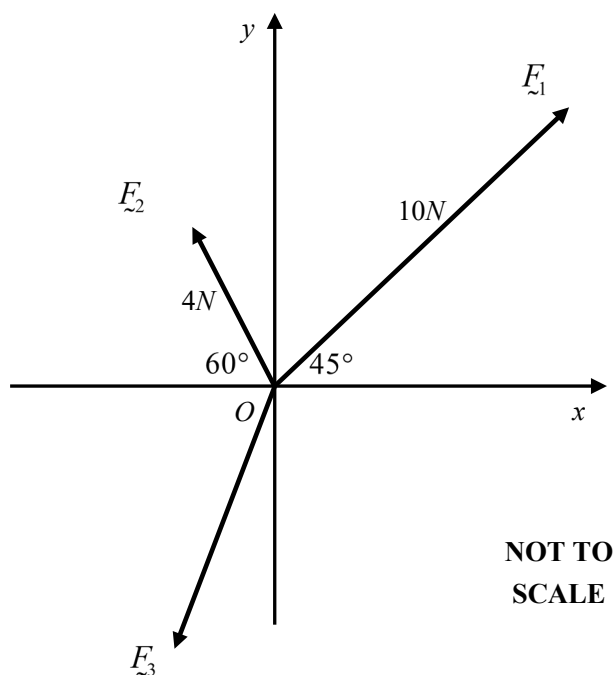


- (i) Write down the two vectors, in terms of $\underline{a}$ and $\underline{b}$, representing the diagonals of the parallelogram. **1**
- (ii) Using dot products, show that the diagonals of a parallelogram are perpendicular when $|\underline{a}| = |\underline{b}|$. **2**

Question 7 (Continued)

- (b) Three forces act simultaneously on a particle situated at the origin as shown in the diagram. **Force 1** ($\vec{F}_1$) has a magnitude of $10N$ and direction of 45° above the horizontal in the first quadrant and **Force 2** ($\vec{F}_2$) has a magnitude of $4N$ and a direction of 60° above the horizontal in the second quadrant.

The third force, $\vec{F}_3$, is also applied simultaneously so that the particle does **not** move.



Find $\vec{F}_3$ in the form $\vec{F}_3 = x\vec{i} + y\vec{j}$.

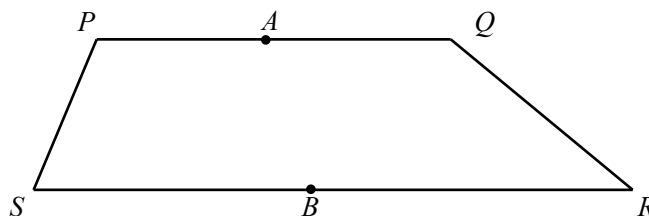
3

End of Question 7.

Question 8 (7 marks) Use a SEPARATE writing booklet

(a) Find the term independent of x in the expansion of $\left(2x^2 - \frac{3}{x}\right)^9$. **3**

(b) $PQRS$ is a trapezium with A and B being the midpoints PQ and SR respectively.



Let $\overrightarrow{PA} = \underline{a}$ and $\overrightarrow{SB} = \underline{b}$

(i) Show that $\overrightarrow{AB} = \underline{a} - \underline{b} + \overrightarrow{QR}$ **2**

(ii) Hence or otherwise show that $\overrightarrow{AB} = \frac{1}{2}(\overrightarrow{PS} + \overrightarrow{QR})$ **2**

End of Question 8.

Question 9 (7 marks) Use a SEPARATE writing booklet

- (a) By considering $(1+x)^{n+3} = (1+x)^n (1+x)^3$ show that: **3**

$$\binom{n+3}{k} = \binom{n}{k} + 3\binom{n}{k-1} + 3\binom{n}{k-2} + \binom{n}{k-3}$$

- (b) By using the formula for the binomial theorem $\binom{n}{r} = \frac{n!}{r!(n-r)!}$: **DO NOT PROVE THIS**

(i) Show that $\frac{1}{\binom{n}{r}} = \frac{r}{r-1} \left[\frac{1}{\binom{n-1}{r-1}} - \frac{1}{\binom{n}{r-1}} \right]$, where $r \geq 2$. **2**

- (ii) Hence show that for integer values of m where $m \geq r$.

Show that $\frac{1}{\binom{r}{r}} + \frac{1}{\binom{r+1}{r}} + \frac{1}{\binom{r+2}{r}} + \dots + \frac{1}{\binom{m}{r}} = \frac{r}{r-1} \left[1 - \frac{1}{\binom{m}{r-1}} \right]$ **2**

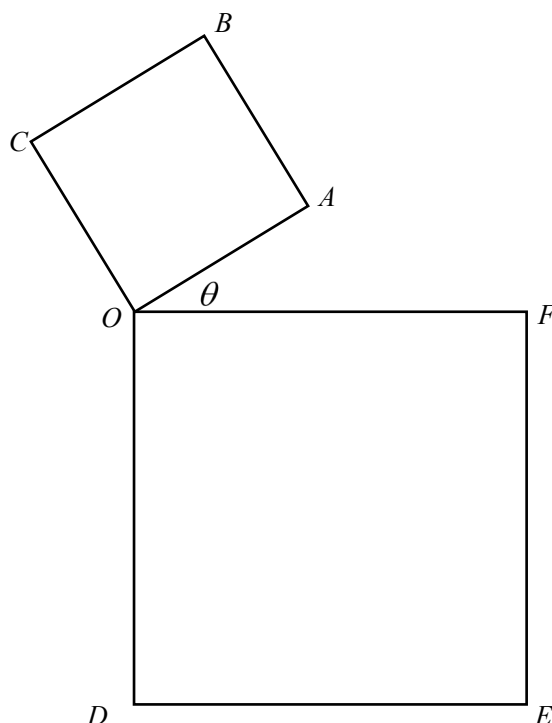
End of Question 9.

Question 10 (7 marks) Use a SEPARATE writing booklet

- (a) By considering the expansion of $(1+x^2)^n$ show that: 3

$$n(2)^n = 2\binom{n}{1} + 4\binom{n}{2} + 6\binom{n}{3} + \dots + 2n\binom{n}{n}.$$

- (b) The Diagram below shows two squares $OABC$ and $ODEF$, with $|\overrightarrow{OA}| = 1$ and $|\overrightarrow{OD}| = 2$
 $\angle AOF = \theta$, where $0 < \theta < \frac{\pi}{2}$. Let $\overrightarrow{OF} = 2\mathbf{i}$, $\overrightarrow{OD} = -2\mathbf{j}$ and $\overrightarrow{OA} = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$



- (i) Show that $\overrightarrow{OC} = -\sin \theta \mathbf{i} + \cos \theta \mathbf{j}$ 2
- (ii) Use vector methods to show that, if the points A , B and E are collinear then: 2

$$\cos \theta - \sin \theta = \frac{1}{2}$$

End of Paper

BLANK PAGE

BLANK PAGE